

Physics 3102 - Quantum Transport

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Introduction:

Notes written in fall 2025 following Sergey Frolov's University of Pittsburgh, Spring 2013 youtube lectures.

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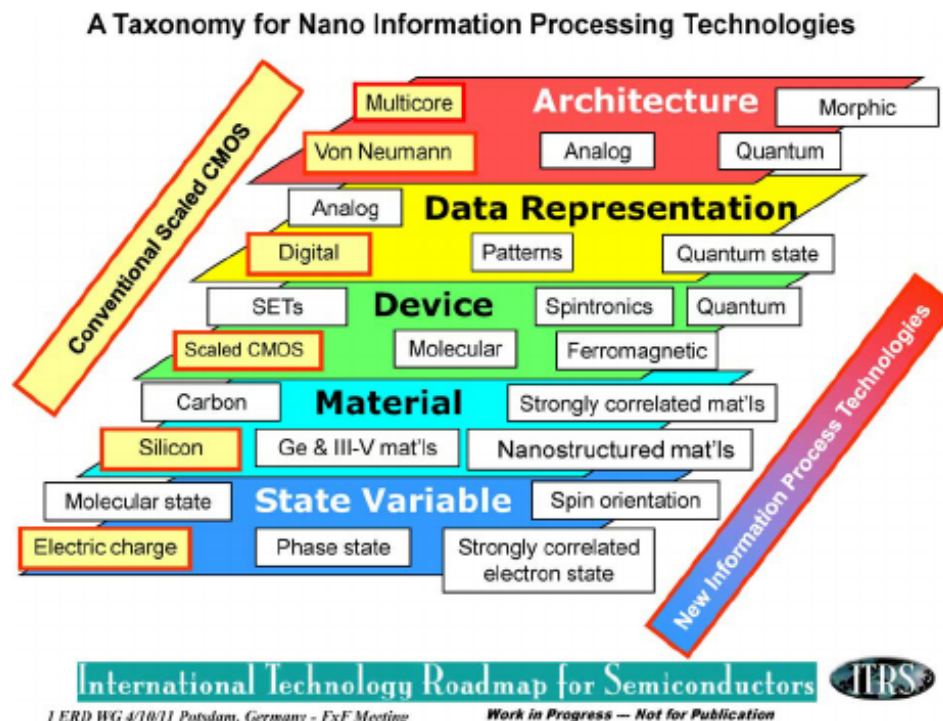
1 Introduction

1.1 Course Overview

- Course Information:
 - Instructor: Sergey Frolov, University of Pittsburgh, Spring 2013
 - Course structure includes homework assignments reviewing scientific papers and a final project proposing a new experiment
 - Experimentalist's perspective on quantum transport

1.2 What is Quantum Transport?

- Quantum transport is the study of how charged particles (like electrons) move through materials and devices when they behave as quantum mechanical objects.
- To get our transport to behave quantum mechanically we can:
 - shrink the size of our device to be comparable to the electron's wavelength (nanoscale devices)
 - cool the device to very low temperatures to reduce thermal vibrations (would want energy scale of wavefunction to be larger than the thermal energy)
- metal-oxide-semiconductor field-effect transistor (MOSFET) cannot be made arbitrarily small because of quantum tunneling through the gate oxide layer (CMOS is the manufacturing process)



1.3 Key Phenomena and Applications

- Key phenomena in quantum transport include:

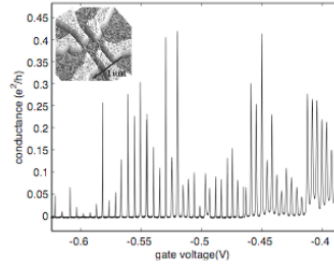


Figure 1.1: Conductance quantization in quantum point contacts showing discrete steps in units of G_0 .

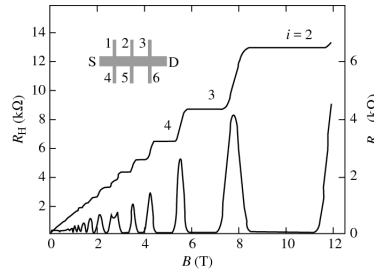


Figure 1.2: Quantum Hall Effect: In a 2D electron gas subjected to a strong perpendicular magnetic field, the Hall resistance becomes quantized in integer multiples of R_0 , leading to precisely defined resistance plateaus.

- Conductance quantization (ballistic transport in low-dimensional systems)
- Quantum interference (e.g., Aharonov-Bohm effect)
- Tunneling and Coulomb blockade
- Mesoscopic superconductivity
- Quantum bits (qubits)
- Topological quantum phases
- Examples of new device concepts:
 - quantum computing (superposition states used for qm algos with speedup over classical algos)
 - spintronics (intrinsic spin used for new logic devices)
 - optoelectronics (device to generate or detect radiation on single photon level)
 - advanced electronics leveraging quantum effects at the nanoscale

1.4 Fundamental Quantum Constants

- Quantum of Charge: $e = 1.6 \times 10^{-19} \text{ C}$
- Quantum of Conductance: $G_0 = \frac{e^2}{h} = 3.87 \times 10^{-5} \Omega^{-1}$
- Quantum of Resistance: $R_0 = \frac{h}{e^2} = 25.8 \text{ k}\Omega$
- Quantum of Magnetic Flux: $\Phi_0 = \frac{h}{e} = 4.14 \times 10^{-15} \text{ webers}$

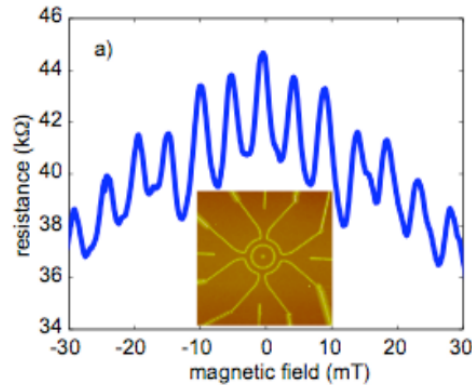


Figure 1.3: The Aharonov-Bohm effect: electrons traveling in a region with zero magnetic field can still be affected by a magnetic flux enclosed by their path, leading to observable phase shifts in their wavefunctions.

1.5 Electron Physics

- Basic electrons: mass m_e , charge e , spin $1/2$
 - PIAB
 - Waves in waveguides
 - Coulomb blockade
 - Zeeman effect
- Electrons + interactions + bandstructure effects:
 - Band structure effects → effective mass like $0.044 m_e$ in GaAs (great that it is lower so that it has higher mobility and easier to get quantum effects) and 0 in graphene
 - electron-electron interaction → fractional quantum Hall effect, Kondo effect (electron with spin on island, conduction electrons in leads screen the spin)
 - Hyperfine coupling → interaction with lattice nuclear spin "bath"
 - Spin-orbit coupling → (also a band structure effect) interaction with electric fields in a lattice, motion of electron through electric field is seen as magnetic field in electron's rest frame, couples to spin of electron (i.e. control of movement of electron can control its spin)
 - superconductivity → pairing of two electrons into (spin singlets called Cooper pairs) a boson flow of supercurrent without resistance

1.6 Landmark Experiments

- The quantum Hall effect (2D): Si-MOSFET (Klitzing et al. 1980)
- Quantum point contacts: gates over 2DEG form 1D channel in semiconductor heterostructure, apply source and drain bias and you measure quantized conductance (Wees et al. 1988)
- Real-world experimental demonstrations (39:40):
 - Single spin, single shot readout: quantum dot in GaAs (Elzerman et al. 2004)
 - Single spin control with magnetic field (Koppens et al. 2006)
 - Coupling between two single spins (Petta et al. 2005)
 - Superconducting Flux Qubit (Mooij et al. 1999) (SQUID does the measurement of the flux qubit)

2 Energy and Length Scales

2.1 Energy Scales

Charging energy (associated with one electron) $E_C = \frac{e^2}{2C}$ (0:51)

- Capacitance of a sphere of radius R: $C = 4\pi\epsilon_0 R$.
- What is the energy needed to charge the sphere with one electron?

Radius R	Capacitance C	Charging Energy E/k_B
$10\mu m$	10^{-15} F	0.84 K
$1\mu m$	10^{-16} F	8.4 K
$0.1\mu m$	10^{-17} F	84 K
$0.01\mu m$	10^{-18} F	840 K

Overview of energy scales:

- $k_B T$: thermal energy scale at room temp is 26 meV (4:39)
- eV: energy an electron gains when crossing a potential difference V (7:22)
- μ chemical potential the energy required to add one particle to the system
- $\mu_B B$: Zeeman energy of an electron in a magnetic field B
- E_F : Fermi energy (it is the energy of the highest occupied electron state); for metals it is on the order of a few eV. For semiconductors it can be much smaller. (9:23)
- ΔE : level spacing, energy difference between single-particle states calculated from the Schrödinger equation (11:23)
- E_C : charging energy (more on this when we cover coulomb blockade)
- E_T : Thouless energy the energy scale of coherence effects (13:11)

2.2 Length Scales in the Quantum Regime

- For mesoscopic and nanoscale structures: device dimensions are comparable to the fundamental size of electron. This causes their properties to be influenced by quantum mechanical effects.
- Size of electron is essentially given by fermi wavelength λ_F (20:40)
 - in most metals, electron density is very large (10^{21}cm^{-3}), and fermi is thus order of a few nanometers.
 - in semiconductors, lower carrier density ($\ll 10^{21}\text{cm}^{-3}$) and thus larger fermi wavelength of several tens of nanometers.

2.3 Scattering and Mean Free Path

- Bloch's theorem says that electrons in a perfect crystal lattice will oscillate without scattering.
- Scattering happens in real materials. electrons scatter from any disorder, such as defects and impurities but also from other electrons and phonons.

- Golden rule in Quantum mechanics says scattering from static potential does not change the energy of the electrons, so scattering off fixed impurity is elastic, and scattering off phonons or other electrons is inelastic.
- Elastic and inelastic scattering length scales are material and temperature dependent.
- Mean free path l is the average distance an electron travels between scattering events (correlation between initial and final momentum is lost – randomization). (34:39)

$$l = v_F \tau$$

when τ is the mean free time between scattering events.

- Mobility: $\mu = \frac{e\tau}{m}$ where τ is the mean free time between scattering events.
- Conductivity: $\sigma = \frac{ne^2\tau}{m} = ne\mu$
- 2D diffusion constant $D = \frac{v_F^2\tau}{2}$
- 1D diffusion constant $D = v_F^2\tau$
- There exists an elastic mean free path, and an inelastic mean free path.

2.4 Coherent vs Incoherent Transport

- Disruption of interference effects from electron phase breaking time τ_ϕ (often don't distinguish from inelastic scattering time τ_i but they are not the same because inelastic scattering does not always cause phase breaking).
- Phase coherence length $L_\phi = \sqrt{D\tau_\phi}$, is avg distance electrons diffuse in the material before their phase is disrupted through scattering. (40:01)
- To observe interference effects, this length must be comparable to device size which requires experiments be performed at low temperatures (to reduce phonon scattering).
- Phase coherence for a small loop of wire at low temperatures with an applied magnetic field can lead to Aharonov-Bohm oscillations in the conductance of the loop as a function of magnetic field. If you make the loop bigger than 100 microns you will not see the oscillations because the phase coherence length is too small.

2.5 Transport Regimes

- Since submicron structure can be fabricated on length scales smaller than the average impurity spacing in semiconductors, we can study different transport regimes:
 - Diffusive transport: electrons scatter multiple times off impurities, phonons, other electrons, etc. Mean free path l much smaller than device length L ($l \ll L$)
 - Quasi-ballistic transport: electrons scatter a few times, $l \sim L$
 - Ballistic transport: electrons travel through the device without scattering, $l \gg L$
- When is transport diffusive/ballistic and classical/quantum?
 - diffusive and classical: $\lambda_F, l_i, l_e \ll L$
 - diffusive and quantum: $\lambda_F, l_e \ll L, l_i$
 - ballistic and classical: $\lambda_F \ll L < l_e, l_i$
 - ballistic and quantum: $\lambda_F, L < l_e < l_i$

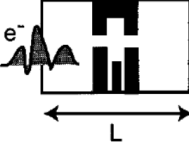
 conventional device:	 mesoscopic device:
$L \gg l_e$ diffusive	$L \lesssim l_e$ ballistic
$L \gg l_\phi$ incoherent	$L \lesssim l_\phi$ phase coherent
$L \gg \lambda_F$ no size quantization	$L \lesssim \lambda_F$ size quantization
$e^2/C \ll k_B \Theta$ no single electron charging	$e^2/C \gtrsim k_B \Theta$ single electron charging effects

Figure 2.1: Different length scales relevant for electron transport in a conductor. The Fermi wavelength λ_F characterizes the quantum nature of electrons, while the mean free path l indicates the average distance an electron travels between scattering events. The phase coherence length L_ϕ represents the distance over which an electron maintains its quantum phase coherence, crucial for observing quantum interference effects.

		GaAs(100)	Si (100)	UNITS
Effective Mass	m	0.067	0.19	$m_e = 9.1 \times 10^{-28} \text{ g}$
Spin Degeneracy	g_s	2	2	
Valley Degeneracy	g_v	1	2	
Dielectric Constant	ϵ	13.1	11.9	$\epsilon_0 = 8.9 \times 10^{-12} \text{ F m}^{-1}$
Density of States	$\rho(E) = g_s g_v (m/2\pi\hbar^2)$	0.28	1.59	$10^{11} \text{ cm}^{-2} \text{ meV}^{-1}$
Electronic Sheet Density ^a	n_s	4	1-10	10^{11} cm^{-2}
Fermi Wave Vector	$k_F = (4\pi n_s / g_s g_v)^{1/2}$	1.58	0.56-1.77	10^6 cm^{-1}
Fermi Velocity	$v_F = \hbar k_F / m$	2.7	0.34-1.1	10^7 cm/s
Fermi Energy	$E_F = (\hbar k_F)^2 / 2m$	14	0.63-6.3	meV
Electron Mobility ^a	μ_e	10^4 - 10^6	10^4	$\text{cm}^2/\text{V} \cdot \text{s}$
Scattering Time	$\tau = m\mu_e/e$	0.38-38	1.1	ps
Diffusion Constant	$D = v_F^2 \tau / 2$	140-14000	6.4-64	cm^2/s
Resistivity	$\rho = (n_s e \mu_e)^{-1}$	1.6-0.016	6.3-0.63	k Ω
Fermi Wavelength	$\lambda_F = 2\pi/k_F$	40	112-35	nm
Mean Free Path	$l = v_F \tau$	10^2 - 10^4	37-118	nm
Phase Coherence Length ^b	$l_\phi = (D\tau_\phi)^{1/2}$	200-...	40-400	$\text{nm}(T/\text{K})^{-1/2}$
Thermal Length	$l_T = (\hbar D / k_B T)^{1/2}$	330-3300	70-220	$\text{nm}(T/\text{K})^{-1/2}$
Cyclotron Radius	$l_{\text{cycl}} = \hbar k_F / eB$	100	37-116	$\text{nm}(B/\text{T})^{-1}$
Magnetic Length	$l_m = (\hbar / eB)^{1/2}$	26	26	$\text{nm}(B/\text{T})^{-1/2}$

2.6 Non-Ohmic Transport Examples

- Current-voltage characteristics (55:13)
- Superconductor-Insulator-Superconductor (SIS) junction (Josephson junction), different than Ohm's law!
 - Current biased: as we increase I , voltage V across junction remains zero until I reaches critical current I_c at which point V jumps to a finite value and then increases linearly with I .
 - Voltage biased: as we increase V , current I across junction remains zero until V reaches critical voltage V_c at which point I jumps to a finite value and then increases linearly with V .
- Quantum point contact "waterfall plot" showing conductance quantization (S. M. Cronenwett et al. 2002)
- Spin blockade in a quantum dot showing non-symmetric current-voltage characteristics due to Pauli exclusion principle (Ono et al. 2002)

3 Materials for Quantum Transport

3.1 Two-Dimensional Electron Gas (2DEG)

- 2D Electron Gas (2DEG) (9:01):
 - Came from industry that was needing faster/smaller devices (note they have large mean free paths i.e. small number of impurities, and high electron mobility).
 - Often in semiconductor heterostructures, formed at interface.
 - Also in Metal-oxide-semiconductor (MOS) structures (14:38).
 - Materials are very well matched at interface so there are few defects.
 - To fill 2DEG you can either apply gate voltage or use doping.
 - Electron mobility goes up as temperature goes down (less phonon scattering).

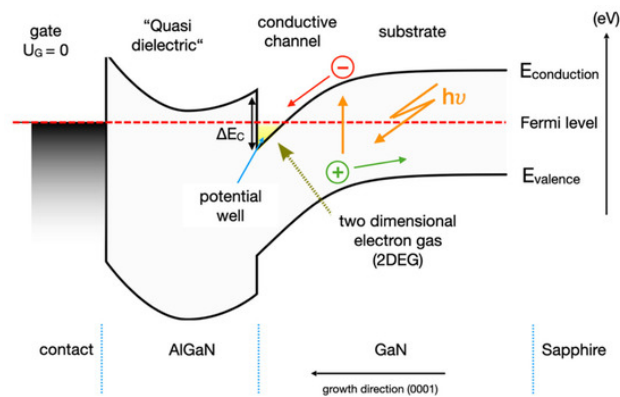


Figure 3.1: 2DEG in a semiconductor heterostructure.

- Place metal on surface and then heat it up (anneal) so that it diffuses through the semiconductor and creates a contact to the 2DEG.
- To make quantum dots we have metals on surface that we apply (very negative) voltage to (gates) to locally deplete the 2DEG.

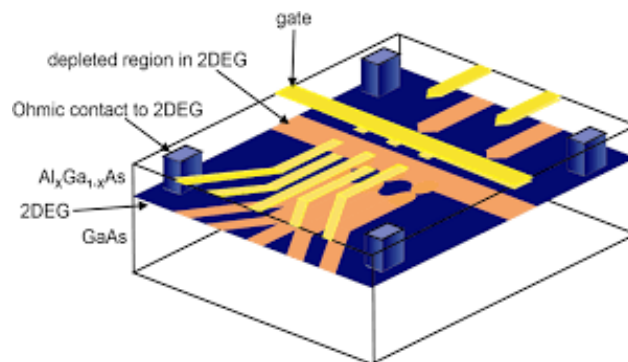


Figure 3.2: Example quantum transport device using a 2DEG.

3.2 One-Dimensional Systems (Nanowires)

- Metallic or semiconductor nanowires (1D systems) (22:50):
 - Bottom-up growth (e.g., VLS process) (26:30)
 - Top-down etching from 2D structures

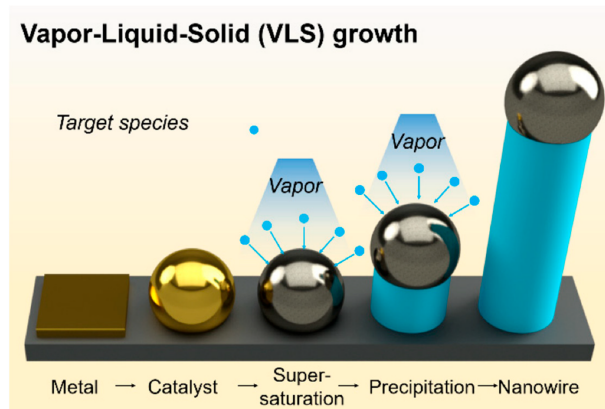


Figure 3.3: VLS grown nanowire process.

3.3 Zero-Dimensional Systems (Quantum Dots)

- Semiconducting quantum dots (0D systems) (26:50):
 - electron motion strongly confined in all 3 dimensions.
 - quantization so that these are like artificial atoms.
 - can be made in 2DEG by applying voltages to gates (top-down).
 - dominant transport is single electron tunneling.
 - can also be by pillars etched out of 2deg and also self-assembled (strained induced) InAs dots (bottom-up) (29:20).
- Metallic clusters & semiconductor dots (0D): made by chemists, synthesized in solution and can make different sizes that can emit different colors.
- Single molecules and chains of atoms
 - two electrodes placed with tiny gap and then deposit molecules and some will bridge the gap.

3.4 Carbon-Based Materials

- Carbon base material-systems: fullerenes (1985, 0D "buckyballs"), carbon nanotubes (1991, 1D) (50:19), graphene (2004, 2D) (48:09).

3.5 Density of States

- Free electrons in three dimensions (40:29):
 - Schrodinger equation: $-\frac{\hbar^2}{2m} \nabla^2 \psi = E\psi$.

- The eigenstates are: $\psi_{\vec{k}}(\vec{r}) = \frac{1}{(2\pi)^3} e^{i\vec{k} \cdot \vec{r}}$ with $\vec{k} = (k_x, k_y, k_z)$.
- The energy eigenvalues are: $E_{\vec{k}} = \frac{\hbar^2 \vec{k}^2}{2m} = \frac{\hbar^2}{2m} (k_x^2, k_y^2, k_z^2)$.
- The unitary volume of the state in k-space is $(2\pi)^3$.
-
- The number of states in a 3D k-space volume $d\vec{k} = dk_x dk_y dk_z$ is given by:

$$g(\vec{k})d\vec{k} = \frac{2}{(2\pi)^3} d\vec{k}$$

where the factor of 2 is for spin degeneracy.

- $d\vec{k}$ can be written as:

$$d\vec{k} = \frac{4}{3} \pi ((k + dk)^3 - k^3) = 4\pi k^2 dk$$

leading to:

$$g(\vec{k})d\vec{k} = \frac{1}{\pi^2} k^2 dk$$

- By calculating k^2 and dk/dE :

$$k^2 = \frac{2mE}{\hbar^2} \quad , \quad \frac{dk}{dE} = \sqrt{\frac{2m}{E\hbar^2}} \frac{1}{2}$$

$$\frac{1}{\pi^2} k^2 dk = \frac{1}{\pi^2} \left(\frac{2mE}{\hbar^2} \right) \frac{dk}{dE} dE = \frac{1}{2\pi^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} \sqrt{E} dE$$

- Which is the density of states in 3D for free electrons:

$$g(E) = \frac{1}{2\pi^2} \left(\frac{2m}{\hbar^2} \right)^{3/2} \sqrt{E}$$

- Can also do this for 2D and 1D and 0D systems:

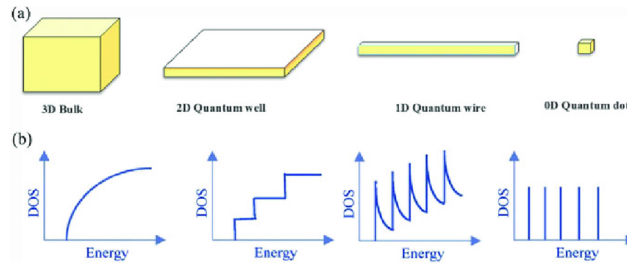


Figure 3.4: Density of states in different dimensions.

3.6 Band Structure

- Band structure and symmetry points (54:18)
- Direct vs indirect band gaps
 - Wider bands are heavier effective mass
 - Narrower bands are lighter effective mass
- Bands vs sub-bands: sub bands based on excited states in quantum wells (not in 3D bulk material)

3.7 Novel Quantum Materials

- Newish quantum transport materials (1:08:29):
 - Bi₂Se₃ topological insulator
 - MoS₂ (transition metal dichalcogenides)
 - LAO/STO (lanthanum aluminate/strontium titanate)

4 Technology

4.1 Overview

- Crystal growth (1:17) (normally done out of house, so that we ensure it is high quality) → nanofabrication (normally done in house because no one wants to do it for you) → Low-T electrical measurements (science!)

4.2 Crystal Growth Techniques

- Molecular Beam Epitaxy (MBE) (4:16):
 - MBE is performed in an ultra-high vacuum (UHV) chamber.
 - A heat source evaporates pure elements, which then condense on a heated substrate.
 - Atoms self-assemble into crystalline layers.
 - The process involves ballistic (of atoms) deposition (molecular flow).
 - Temperature is a critical parameter.
 - Effusion cells are used to carry very pure elements (like Gallium, Arsenic, Aluminum, Indium).
 - A RHEED (reflection high energy electron diffraction) gun reflects electrons off the surface and reads the pattern with a sensor to monitor growth in real time. Signal oscillates, intensity determined by surface roughness.
- Metal Organic Vapor Phase Epitaxy (MOVPE) (or MOCVD) (17:09):
 - Temperature usually 600-1200C
 - Uses metal-organic precursors (like trimethylgallium, triethylgallium, arsine, phosphine) that decompose on the hot substrate to deposit the desired elements (i.e. organics leave and leave semiconductor elements behind).
 - Used to make nanowires
- Pulsed Laser Deposition (PLD):
 - UHV chamber with window
 - laser shoots at target material, ablates material which deposits on substrate
 - used to make complex oxides (LAO), HTSCs (YBCO, BSCO etc.), and also semiconductors

4.3 Nanofabrication Techniques

- Nanofabrication (30:21): pattern definition (photolithography (31:15), Electron beam lithography (39:30), and nano imprint lithography), Thin film deposition (evaporation (31:55), sputtering using plasma), Etching (chemical etching – wet and dry, Ion milling)
- Excimer Laser Stepper: resolution is around 65nm, can make smaller features possible, fast, very expensive.
- Stamping nanostructures: PDSM stamp, nanoimprint, step and flash imprint lithography (around 22nm resolution across large area, fast, need master, overlay is difficult).
- Electron Beam Lithography (EBL): a pattern generator to SEM (beam is scanned across surface), very high resolution (sub 10nm), slow, serial process, need conductive substrate or coating. Several million dollars.

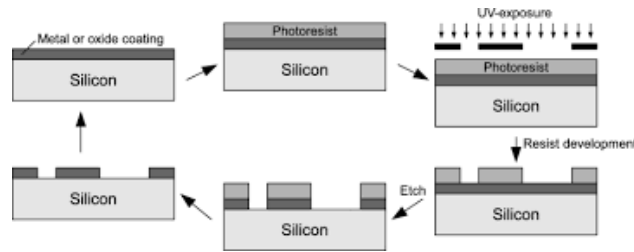


Figure 4.1: photolithography process

- Focused Ion Beam (FIB) (44:09): use atoms like Ga, to mill away material. Pretty similar properties to EBL.
- Nanofabrication journey:
 - you don't get unlimited good semiconductor wafers
 - find defects and figure out if you can avoid them in your device design.
 - practice with just GaAs at first without the 2DEG

4.4 Low Temperature Technology

- Low-T technology (50:51):
 - Liquid helium 4 at 4.2K (bath cryostat) (51:50).
 - Pumped helium 4 – 1.5K (note that superfluid transition is at 2.17K).
 - Helium 3 – 0.3K (expensive, rare, hard to handle) (55:56)
 - Dilution Refrigerator (mix of helium 3 and helium 4) – 0.01K (58:55)
 - Adiabatic demagnetization refrigerator – 0.001K, use solids and apply magnetic fields to cool down (entropy increases, temperature goes down), no helium 3 required. (1:07:24)
- Cooling electrons down (1:11:46): electron temperature is not the same thing as the lattice temperature (phonons) i.e. the temp of the fridge.
 - Let electrons have time to cool down by wrapping wires around stages of fridge. Electron-phonon coupling. Add resistor to cool down electrons.
 - Need to filter out high frequency noise (RF) from room temperature electronics. Use RC filters, copper powder filters, pi-filters, etc.

5 Ballistic Transport

5.1 Transport Regimes

- Diffusive transport: electrons scatter multiple times off impurities, phonons, other electrons, etc. Mean free path l much smaller than device length L ($l \ll L$) (0:09)
- Quasi-ballistic transport: electrons scatter a few times, $l \sim L$
- Ballistic transport: electrons travel through the device without scattering, $l \gg L$

5.2 Drude Model and Diffusive Transport

- The Drude model describes diffusive transport (1:31), based on the free electron gas picture. Electrical resistance comes from scattering (information is lost when scattering).
- This gives us equation for velocity of electron in an electric field and mobility of electron, and conductivity.

$$\begin{aligned}\vec{v}_d &= -\frac{e\vec{E}\tau}{m} \\ \mu_d &= \frac{e\tau}{m} \\ \sigma &= \frac{ne^2\tau}{m} = ne\mu_d\end{aligned}$$

- Einstein relation: electrons near E_F diffuse due to density gradient.
- Ohms law and drude model are in diffusive regime

5.3 Ballistic Regime

- In the ballistic regime, electrons travel through the device without scattering. The resistance comes from the contacts (interfaces) to the device, not from within the device itself.
- With non-local measurement (voltage is not measured along the current path to electrical ground) you can measure electron focussing in a magnetic field (8:15) (famous non-local measurement is quantum hall effect).

5.4 Quantum Point Contacts

- Split gate hetero structure in 2DEG GaAs/AlGaAs, apply negative voltage to deplete 2DEG under gates and create a narrow constriction (quantum point contact, QPC). When you make the constriction narrow enough you get quantized conductance in units of $G_0 = \frac{2e^2}{h}$ (16:37).
- Constricts to 1D transport, so electrons can only move forward or backward. The conductance is given by the Landauer-Buttiker formula (17:31).

5.5 Temperature and Field Effects on QPCs

- QPC are not perfect, if you increase temp the plateaus get washed away (21:41). For finite temp we must consider a finite occupation of the energy levels.
- The width of the derivative of the fermi function is $\sim 3.5k_B T$. If $3.5k_B T$ is comparable to the sub-band spacing then the plateaus get washed out.
- Transmission resonance can also wash out the plateaus.
- Reflection can be from impurities or Backscattering from defects can obscure quantization (26:41). Tapered/smooth and short potential constriction is better for transmission closer to $T = 1$ and $R = 0$.
- Can measure in nanowires also but you need very high magnetic fields to see quantized conductance.
- Perpendicular magnetic field also makes plateaus more stretched out in gate voltage and can suppress backscattering (29:32).
- Parallel magnetic field splits spin degeneracy and you get $1e^2/h$ plateaus.

5.6 Advanced Techniques

- Scanning gate microscopy: technique to image electron trajectories and interference patterns in 2D electron gases (35:53).

6 Quantum Point Contacts II

6.1 GaAs/AlGaAs Heterostructures and QPCs

- GaAs/AlGaAs heterostructure has electrons that are confined to a 2D layer in the interface between the two materials (0:30). The electrons can move freely in the 2D plane but are confined in the third dimension. These electrons have very high mobility! with mean free path around 25 microns.
- Use gates to apply electric fields to deplete the 2DEG under the gates and create a narrow constriction (quantum point contact, QPC). When you make the constriction narrow enough you get quantized conductance.
- These are modes in the solution to the Schrodinger equation in the constriction. Each mode contributes $2e^2/h$ to the conductance (factor of 2 from spin degeneracy).
- First two papers on this topic are (Wees et al. 1988) and (Wharam et al. 1988).
- See also (Sara M. Cronenwett 2001) for more interesting plots like the waterfall plot.

6.2 Conductance Plateaus and Anomalies

- It shows conductance plateaus which become spin resolved if you apply a magnetic field.
- At 0 B you can also get 1/2 plateaus with high bias (3:35) (but this is not spin-resolved).
- 0.7 or 0.8 plateaus have been debated over for several decades. Likely from interaction effects. When you apply large field it goes to 0.5 plateau.

6.3 Landauer-Büttiker Formalism

- Adiabatic Quantum Transport (27:00): Many modes, some closed some open. Smoothly varying potential. No scattering between modes. This is called the Landauer Buttiker formulism of Quantum Transport.
- Transmission and Reflection of modes.
- Incoming amplitudes $\vec{a}$ and outgoing amplitudes $\vec{b}$.

$$\vec{b} = \hat{s}\vec{a}$$

$$\begin{bmatrix} \vec{b}_L \\ \vec{b}_R \end{bmatrix} = \begin{bmatrix} \hat{r} & \hat{t}' \\ \hat{t} & \hat{r}' \end{bmatrix} \begin{bmatrix} \vec{a}_L \\ \vec{a}_R \end{bmatrix}$$

- Unitarity of S-matrix: $\hat{s}^\dagger \hat{s} = \hat{I}$. Express conservation of the number of particles.
- Fundamental relation is also the Onsager Relation, for symmetry with respect to the time reversal:

$$s_{\alpha n; \beta m}(B) = s_{\beta m; \alpha n}(-B)$$

6.4 Alternative 2DEG Systems

- You can inject a 2DEG by adding electrons to surface of liquid helium (46:51). Lower density electrons, less screening, much stronger Coulomb interactions. If you lower the temperature enough you can get a Wigner crystal (held together in crystal lattice form by Coulomb repulsion). Harder to create quantum confinement because these are real electrons (not effective) here have larger mass of 1 (compared to GaAs 2DEG electrons of 0.067) and thus much smaller wavelength.
- Groups have tried to make QPCs in this system but it is very challenging.

6.5 Spin Effects in QPCs

- Spin effects of QPCs (56:17), apply magnetic field in plane (no orbital effects), adds Zeeman energy to electrons in the 2DEG. Extra plateaus show in conductance at 0.5, 1.5, 2.5, etc. $2e^2/h$.
- The factor of 2 normally comes from spin degeneracy. The magnetic field breaks this degeneracy and splits the modes.
- Gate voltage at odd integer multiples of e^2/h only allows electrons with spins oriented along the field to go through the QPC. This is a spin filter (or injector)!
- Non local spin valve (1:02:59): inject spins at one QPC and detect them at another QPC.

7 Coulomb Blockade

7.1 Introduction to Charging Effects

- Charging effects (0:18)
- Occur in bulk metal.
- Charge quantization effects only observable in small volume well separated from its environment.
- The total capacitance of the island (to the ground, leads, etc):

$$C_{\Sigma} = C + C_{\text{self}} + C_{\text{res}}$$

- With isolated island: charge is quantized, $Q = ne$.
- Energy to add an electron to the island:

$$E = E_c n^2$$

$$E_c = \frac{e^2}{2C_{\Sigma}}$$

7.2 Capacitance and Energy Scales

- If charging energy is not available from external voltage sources, T, etc, then transport is blocked (3:41).
- Capacitances: isolated sphere $C = \epsilon_0 \epsilon_r 2\pi d$, isolated disk $C = \epsilon_0 \epsilon_r 4d$, parallel plate $C = \epsilon_0 \epsilon_r A/d$, nanotube with diameter, r, above a ground plane at distance h $C = \frac{2\pi\epsilon_0\epsilon_r L}{\ln(2h/r)}$.
- Quick estimate: for capacitance per unit length $C' = \epsilon_0 \epsilon_r = \epsilon_r 10 \text{ aF}/\mu\text{m}$
- Mesoscopic systems, typically C in 1-100 fF range. E_c in 0.5-50 meV range (4:09).
- What is the energy needed to charge the sphere with one electron? Energy of charged capacitor:
 $E = qV/2 = q^2/2C$

R	C	E/k_B	E
10 μm	$1.1 \times 10^{-15} \text{ F}$	0.84 K (^3He)	70 μeV
1 μm	$1.1 \times 10^{-16} \text{ F}$	8.4 K (LHe)	0.7 meV
0.1 μm	$1.1 \times 10^{-17} \text{ F}$	84 K (LN ₂)	7 meV
0.01 μm	$1.1 \times 10^{-18} \text{ F}$	840 K (spg)	70 meV

- Need $E_c > kT$ in order to observe charging effects.

7.3 Conditions for Coulomb Blockade

- Conditions for observing Coulomb blockade (12:30):
 - $\Delta t = RC$ is typical time to charge the island, then Heisenberg uncertainty relation: $\Delta E \Delta t = (e^2/C)RC > \hbar$
 - $R \gg \frac{\hbar}{e^2} \approx 25.8 k\Omega$
 - Requirement on temperature/voltage: $kT, eV < E_c$
 - Typical experimental setup: quantum dot connected to source and drain via tunnel barriers, capacitively coupled to gate.

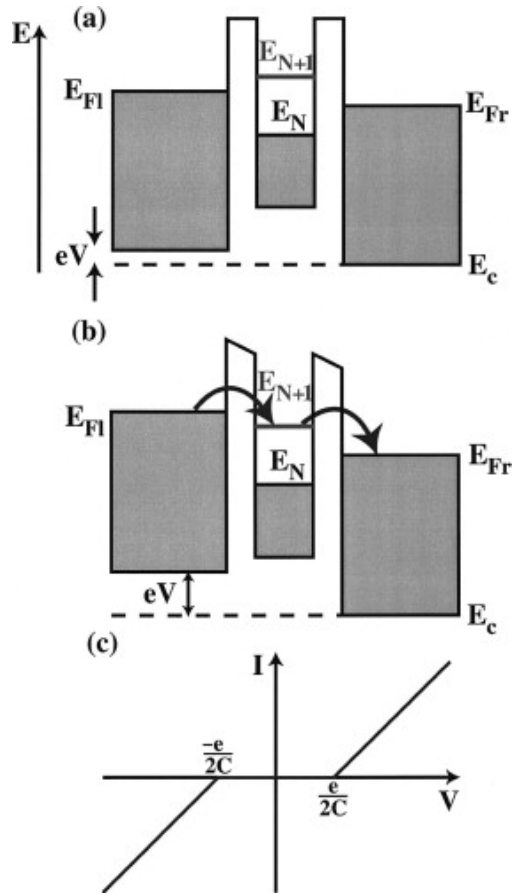


Figure 7.1: Coulomb blockade.

7.4 Single Electron Transistor (SET)

- Tunnel Junction: two conductors separated by thin insulator. Electrons can tunnel through the insulator. Ohms law does not apply.
- Island: single electron box, constant interaction model. no current flows. island potential is given by charge on island plus induced charge from gate.
- We can add electrons one by one to a single electron box by changing the gate voltage.
- Double barrier circuit: single electron transistor (SET).
- Chemical potential is the energy an electrons needs to have in order to enter the system. $\mu(N) = U(N) - U(N-1) = (N-1/2)e^2/C - eV_{\text{ext}}$.
- Current flows if a dot chemical potential lies inside the bias window (window between source and drain chemical potentials) (24:43, 28:03).

7.5 Historical Development and Fabrication

- First measurements of the Coulomb Effect: I. Giaever and H.R. Zeller, Phys. Rev. Lett. 20, 1504 (1968).

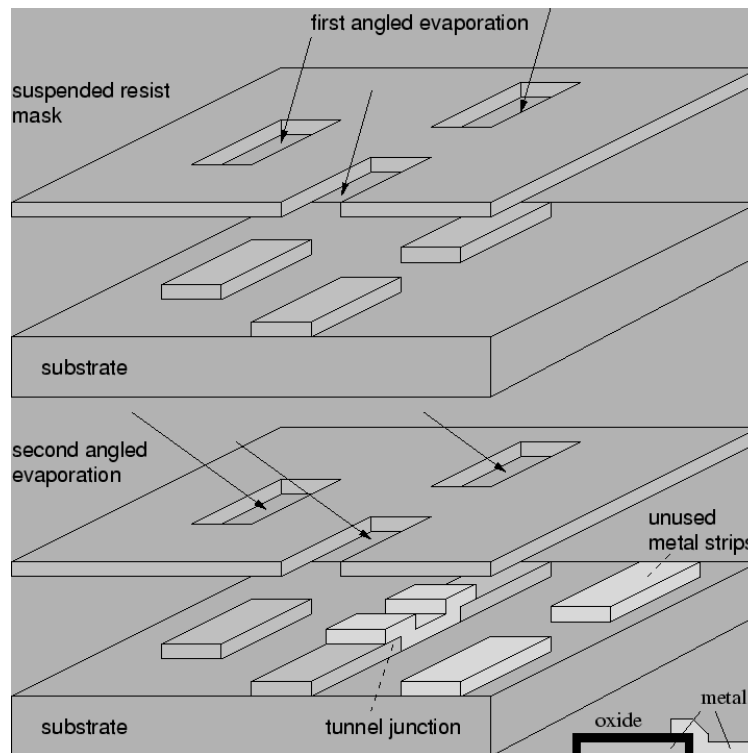


Figure 7.2: Shadow evaporation.

- 20 years later: Observation of Single-Electron Charging Effects in Small Junction by T.A. Fulton and G.J. Dolan, Phys. Rev. Lett. 59, 109 (1987).
- Aside: Shadow Evaporation: (last lecture 2DEG and apply gates to apply potential barriers). But in metals, we can use shadow evaporation to create small tunnel junctions.
- This technique is used for SETs, SQUIDs, Quantum Dots with tunnel barriers, other devices with small tunnel junctions.
- Steps:
 - Evaporate metal onto insulator.
 - Apply resist mask with undercut.
 - Evaporate metal at angle 1.
 - Oxidize to form tunnel barrier.
 - Evaporate metal at angle 2.
 - Lift off resist.
 - Result: two metal layers separated by thin oxide layer.

7.6 Coulomb Diamonds and Stability Diagrams

- Coulomb diamonds: the lines define a region in which there is no current. This region is called the coulomb diamond. At zero bias, current flows at the blue degeneracy points (34:51).

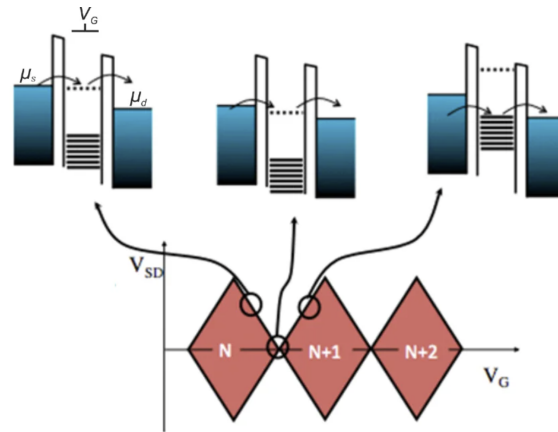


Figure 7.3: Charge stability diagram also known as Coulomb diamonds.

- Gates are even used now to even shift the latter of the energy levels in the dot on top of also the source and drain gates. So that is how we get the Source/Drain gate voltage vs Gate voltage (on the dot) Coulomb diamond plot.
- At the degeneracy points, the energy cost for N and $N+1$ electrons is the same.
- On the lines, either the source or drain chemical potential is aligned with a dot chemical potential.
- Inside diamond, no current flows
- Outside diamond, current flows.
- Slope of lines tells you about the coupling capacitance between the source/drain/gate and the island (37:05).
- You can also figure out the charging energy of the capacitor by looking how stretched from a square diamond it is (compare width and height from centre of diamond). Conversion factor to figure out how much gate voltage needs to be applied to overcome the charging energy.
- Gate traces and stability diagrams are used to characterize quantum dots. Inside the coulomb diamonds, the number of electrons on the island is fixed and no current flows. Outside the Coulomb islands, the number of electrons fluctuates and current flows.
- A low-bias gate trace shows peaks in conductance each time the number of electrons on the dot changes by one (41:21).
- Asymmetric coupling: Coulomb staircase (Matsumoto et al. 1996)
- Paper example with coulomb diamond (Thelander et al. 2003)

7.7 Temperature Effects and Thermometry

- Temperature effects: as temperature increases, the peaks become oscillations until they eventually disappear.
- FWHM of a conductance peak can be used to estimate the electron temperature in the device. There is a quantum and classical equation for this (51:12).
- Remember: electron temperature does not need to equal the fridge temperature (insufficient electron-phonon coupling). Must use wire design and filtering to reduce noise and heating and cool electrons.

7.8 Charge Sensing

- Paper shows off-set charges in coulomb diamond diagram (Bolotin et al. 2004)
- Can get random sudden offset charges in the environment that shift the coulomb diamonds.
- This paves the way to charge sensing.
- QPC (or another QD or SET) can be used as detector for charge in quantum dot.
- Tune QPC to slope between conductance plateaus. Change in charge on dot changes potential landscape and hence conductance of QPC (59:38).

7.9 Applications of SETs

- SET applications: sensitive charge ($10^{-5}e/\sqrt{\text{Hz}}$), readout of charge qubits, readout of spin qubits (via spin-to-charge conversion), current standard (turnstile)
- Single-electron logic and memory devices (Korotkov 1999). Issues: random offset charges, and room temp operation.
- Single-electron turnstile: AC pulse gates at fixed frequency to get current standard $I = ef$. NIST standard: 1 part in 10^8 which is better than what the turnstiles can do (1:02:26).
- Room temp silicon SET (coulomb diamonds not very reproducible so can't be used in our normal computers) (Zhang et al. 1998)

7.10 Cotunneling

- Cotunneling (2nd order perturbation theory): electron tunnels from source to higher level in dot, then to drain (1:10:00).
- Excited state spectroscopy in carbon nanotube double quantum dots (Sapmaz et al. 2005)
- Cotunneling spectroscopy in GaAs quantum dots (Zumbühl et al. 2004)

8 Quantum Dots

8.1 Introduction and Basic Concepts

- Quantum dots differ from single-electron transistors (SETs) due to their quantum confinement energy (0:30).
- Key features: quantized level spacing from quantum confinement, discrete energy levels.
- Requirements: relatively low number of electrons (semiconductors), small island.
- Can be built on different scales:
 - Single molecule quantum dots (nm scale)
 - Self-assembled quantum dots
 - Gate-defined GaAs quantum dots
 - Carbon nanotubes (micron scale)
 - Graphene quantum dots

8.2 Quantum Dots vs Classical SETs

- Few-electron quantum dots in nanowires (Björk et al. 2004)
- Classical dots: regular spacing of conductance peaks (4:27).
- Quantum dots: irregular peak spacing due to additional orbital energy.
- Addition energy includes both charging energy and quantum (orbital) energy (6:30).
- Current-voltage characteristics (10:48):
 - Classical SET: zero current until threshold voltage, then linear current flow (because of continuous density of states)
 - Quantum dot: step-like IV curve reflecting discrete level structure (because of discrete density of states)

8.3 Transport Properties

- Resonant transmission in quantum dots can achieve near-unity conductance when barriers are symmetric (11:14).
- Coulomb diamonds and shell filling (16:00):

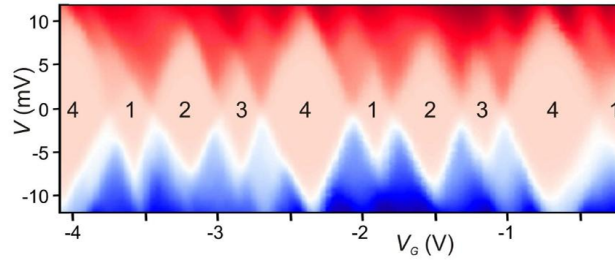


Figure 8.2: Coulomb diamonds in a single-wall nanotube quantum dot (Sapmaz et al. 2005).

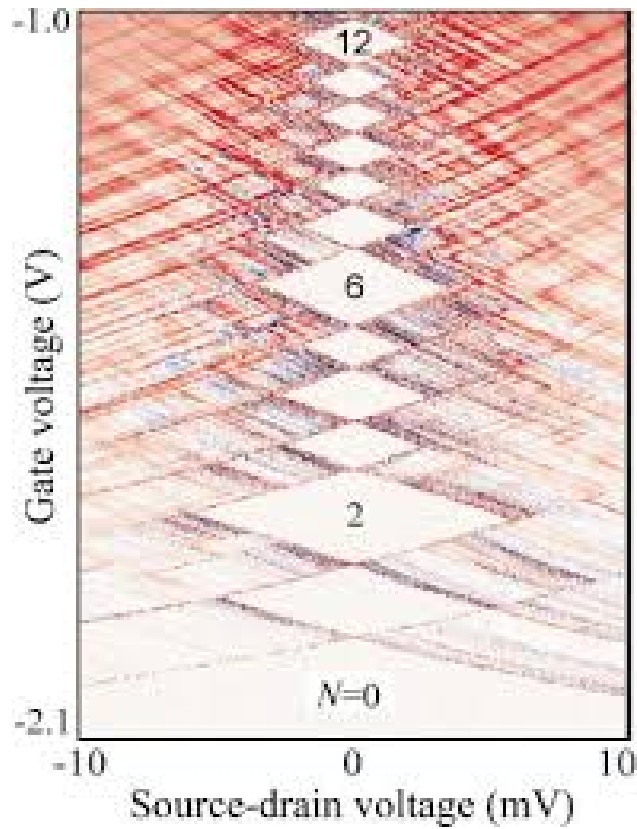


Figure 8.1: Coulomb diamonds illustrating shell filling in quantum dots (L. P. Kouwenhoven et al. 1997).

- Electrons occupy discrete energy levels
- Behavior similar to artificial atoms.
- Characteristic patterns in stability diagrams
- Another example of Coulomb diamonds in a single-wall nanotube quantum dot (20:45).
 - 4 fold shell filling pattern due to spin and valley degeneracy (see 8.2).
- Another example of Coulomb diamonds in a single electron-hole quantum dot (23:05).

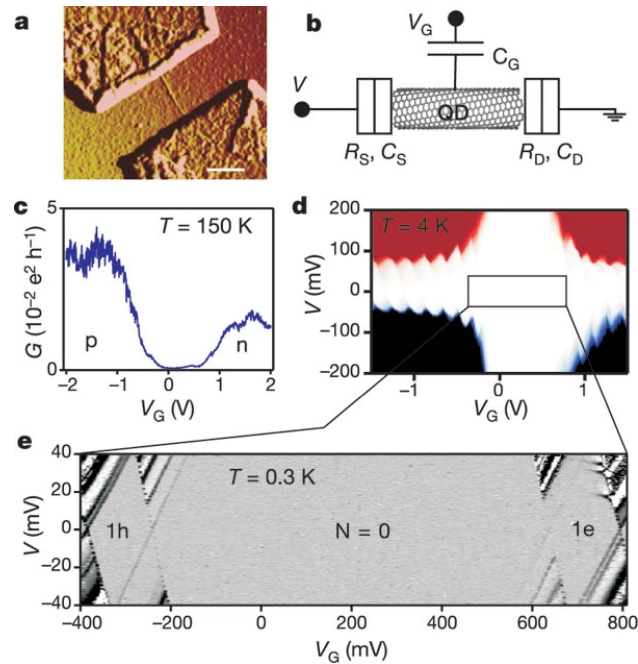


Figure 8.3: Coulomb diamonds in a single electron-hole quantum dot (Jarillo-Herrero et al. 2004).

8.4 Magnetic Field Effects

- Effect of magnetic fields on quantum dot energy levels (25:19):
 - Zeeman splitting
 - Fock-Darwin spectrum

8.5 Level Spectroscopy: Excited States

- Excited states analysis using higher bias measurements (35:55).
- Can reveal:
 - Orbital states
 - Spin states
 - Vibrational states: e.g. vibration assisted tunnelling in a C60 transistor (Park et al. 2000)

8.6 Double Quantum Dots

- Electrostatics of double quantum dots (Wiel et al. 2002; Hanson, L. P. Kouwenhoven, et al. 2007) see 8.6.
- Honeycomb charge stability diagrams (56:00) see 8.7.
 - Now conduction only happens at Triple points: points where three charge states are degenerate (Pothier et al. 1992)

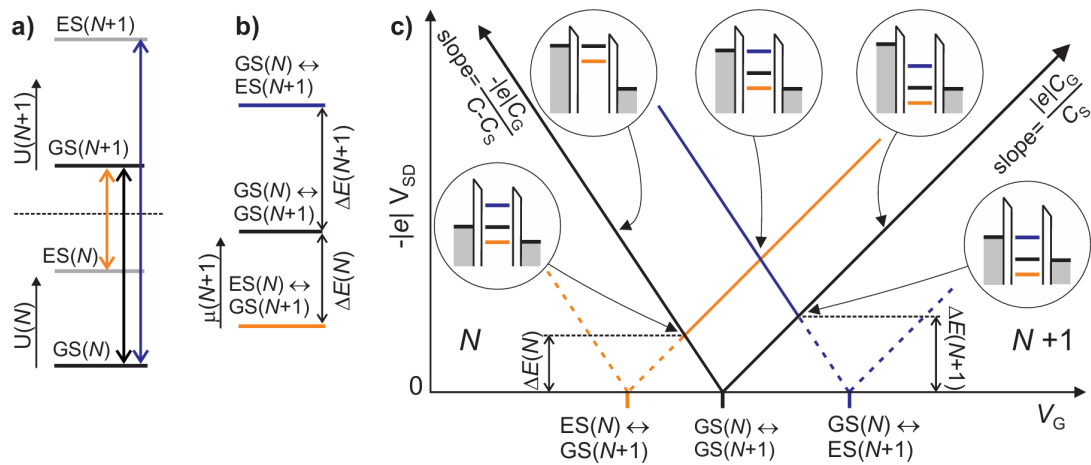


Figure 8.4: Excited states in a quantum dot (Hanson, L. P. Kouwenhoven, et al. 2007).

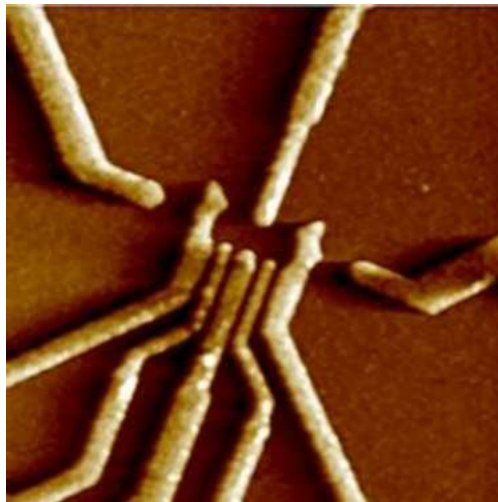


Figure 8.5: SEM image of a double quantum dot device.

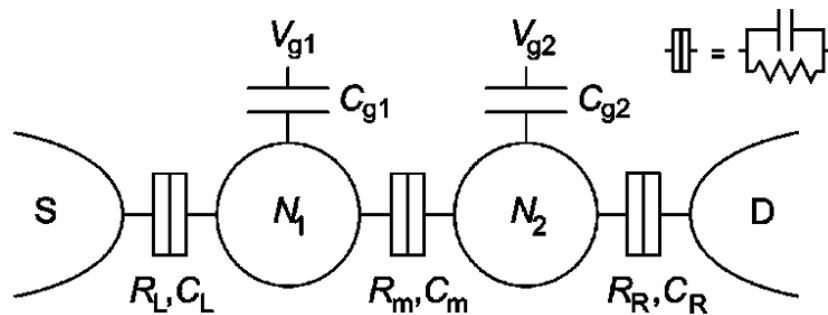


Figure 8.6: Schematic of a double quantum dot system (Wiel et al. 2002)

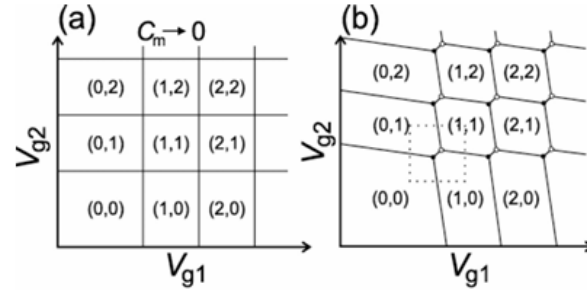


Figure 8.7: Honeycomb charge stability diagram of a double quantum dot (Wiel et al. 2002).

- You can use the Stoof and Nazarov model to extract tunnel rates and inter-dot coupling from these triangles (Stoof et al. 1996)

$$I(\epsilon) = \frac{e}{h} \frac{\Gamma_R |T_{LR}|^2}{\epsilon^2 + \Gamma_R^2/4 + |T_{LR}|^2 (2 + \Gamma_R/\Gamma_L)} \quad (8.1)$$

- DQDs enable study of:
 - Distinguishing from single quantum dot, here we have two ladders of discrete quantum states and the tunnel coupling is only determined by the inner transition without the effects of the outer thermally broadened leads (L. Kouwenhoven 1995; Vaart et al. 1995)
 - Much more sensitive to very narrow transitions
 - Co-tunneling processes
 - Finite bias triangles: regions of allowed transport under finite bias (1:05:19)
- Can also make triple quantum dots but then the stability diagram becomes 3D (Schröder et al. 2007).

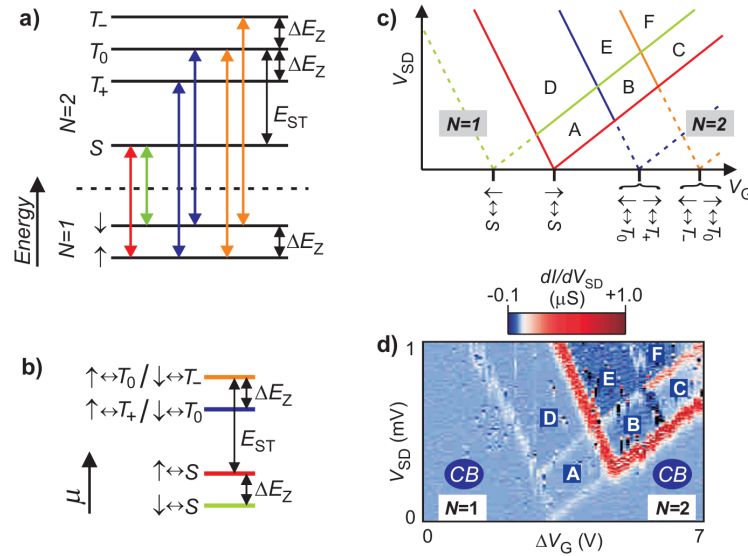


Figure 9.1: Spin states for one and two electrons in a quantum dot (Hanson, L. P. Kouwenhoven, et al. 2007).

9 Spin States in Quantum Dots

This lecture follows the review article by Hanson, L. P. Kouwenhoven, et al. 2007.

- Single Quantum Dots:
 - Zeeman splitting of spin states in magnetic field.
 - Spin Filtering
 - Kondo Effect
- Double Quantum Dots:
 - Singlet and Triplet States
 - Pauli Spin Blockade
 - Spin-orbit and Hyperfine interactions
- $N = 1$ excitation spectrum using CSD, you can see jump from ground state to excited state.
- You can also see even more states when you apply a magnetic field, which gives Zeeman splitting of spin states (Hanson, Witkamp, et al. 2003).
- Let's look at spin states in this figure 9.1.
- Can make a spin filter which only allows one spin orientation to pass through. Theory (Recher et al. 2000) and experiment (Folk et al. 2003)
- Kondo Effect: spin in a quantum dot is 'screened' by spins of electrons in the leads. Electrons in the leads form a 'Kondo Cloud' of opposite spin.
- This is destroyed by finite bias, magnetic field and temperature.

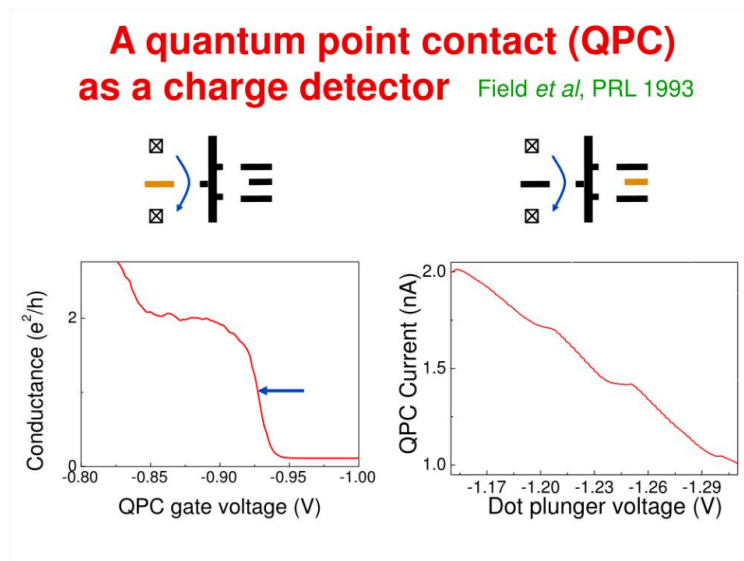


Figure 9.2: QPC as a charge sensor for a quantum dot

- This manifests in QDs as odd fillings having a conductance peak at zero bias via a co-tunneling process (Goldhaber-Gordon et al. 1998; Sara M. Cronenwett et al. 1998).
- Example of Kondo effect using InAs nanowire quantum dot (Kretinin et al. 2011)
- If you apply a magnetic field, you can see the Zeeman splitting of the Kondo peak away from zero bias (Sara M. Cronenwett et al. 1998).
- Now we move to double quantum dots (DQD).
- QPC detector for double dot. The QPC is a tunneling junction between the two dots. QPC detects all charge transitions, also between dots.
- Charge stability diagram of DQD. Labels (N_L , N_R) refer to number of electrons in left and right dot respectively.
- Bias through QPC instead of through the dots.
- The group of Lieven Vandersypen at Delft University of Technology has developed an automated algorithm to find the single electron regime using charge sensors (QPCs) (37:12).
- Tune gate voltage on QPC until you're at final sensitive drop in conductance (Field et al. 1993), see example in figure 9.2.
- When you add electron to dot, the conductance of the QPC shifts, so that we see a plateau. We can always see the last electron added or removed from the dot using the QPC.
- Comparitively the conductance peaks become very small when you reach the last electron in the dot and are hard to see.
- Pauli Spin Blockade (Ono et al. 2002):
 - Consider two electrons in a double quantum dot.
 - If the electrons are in a triplet state, they cannot both occupy the same dot due to Pauli exclusion principle.

- This results in a blockade of current through the double dot when trying to move from (1,1) to (0,2) charge configuration.
- This effect can be used to read out spin states in quantum dots.
- Triplet states: $S = 1$, example: $T_+ = \uparrow\uparrow$, $T_- = \downarrow\downarrow$, $T_0 = \frac{1}{\sqrt{2}}(\uparrow\downarrow + \downarrow\uparrow)$
- singlet states: $S = 0$, example: $S = \frac{1}{\sqrt{2}}(\uparrow\downarrow - \downarrow\uparrow)$
- By biasing a quantum dot, electrons can flow until the current stops due to the spin blockade. This indicates that a spin triplet state has been formed, which can be used to set the initial state of a qubit (50:57).
- Charge stability diagram showing Pauli spin blockade (Nadj-Perge et al. 2010).
- Microwave photon spectroscopy in quantum dots to see the transitions (Schreiber et al. 2011)
- Electron spins in semiconductors:
 - Spin-Orbit interaction
 - Hyperfine interaction
 - Spin-orbit + phonons = spin relaxation
 - Dephasing (10-100ns timescale)
- Probe spin mixing in the spin blockade regime.
 - Hyperfine-induced spin mixing: mixes singlets and triplets near $B=0$. (nuclear spins have no energy at $B=0$ no energy difference between S and T).
 - Spin-orbit induced spin mixing: increases with B field (large energy difference = faster relaxation). Note that spin-orbit has no effect at $B=0$ since time-reversal symmetry is preserved.
- Nuclear spin bath dephases electron spin states via hyperfine interaction.

$$\mathcal{H} = g\mu_B \vec{B} \vec{S} + \vec{S} \sum_i A_i \vec{I}_i \quad (9.1)$$

where A_i is the hyperfine coupling constant for nucleus i , and $\vec{I}_i$ is the nuclear spin operator.

- See (Khaetskii et al. 2002; Merkulov et al. 2002)

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